FRANMARINE UNDERWATER SERVICES PTY LTD

VESSEL START-UP AND SHUT-DOWN

Procedure

MV ROHANI | UVI: 457394

Marine Order 504 (2024) — Schedule 1 Compliance

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Review Frequency

This document shall be reviewed at least every 12 months, or sooner following a significant incident, change of operations, vessel modification, or change in applicable legislation.

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TABLE OF CONTENTS

1. PURPOSE AND SCOPE

This procedure establishes the engine start-up, post-start verification, and shut-down procedure for MV Rohani, fitted with twin Detroit Diesel 6V-92T diesel inboard engines in an enclosed engine room.

1.1 Regulatory Reference

Marine Order 504 (2024), Schedule 1, Clause 11(5)(a).

1.2 Battery System

MV Rohani is fitted with a bank of six batteries arranged in three groups — Port engine, Starboard engine, and House — each group with its own isolator switch. An emergency parallel isolator allows a healthy group to be bridged across for engine starting if a start group is flat. Under normal operations each group is isolated to its own circuit and the emergency parallel isolator remains OFF.

2. POST-LAUNCH CHECKS

Immediately after launching the vessel (or after arriving at the vessel if berthed afloat), the following checks shall be completed before starting engines:

- Confirm the vessel sits well in the water with no unusual list or trim.
- Inspect the vessel for signs of leaks. Check around hull fittings, transom, and bilge area.
- Confirm the bilge system is functioning. MV Rohani is fitted with 3 power-driven bilge pumps, one each in the forward, aft, and engine-room compartments, each on an automatic float switch with manual override. Operate each pump in manual to verify discharge and confirm the float switches are free.

3. ENGINE START-UP PROCEDURE

3.1 Battery Isolation

- Switch the Port, Starboard, and House battery group isolators to ON.
- Confirm the emergency parallel isolator is OFF.
- If an engine-start group is known to be flat, close the emergency parallel isolator to bridge from a healthy group for starting, then open it once the engines are running and charging.

3.2 Pre-Start

- Confirm engine emergency stops are functional and engine-room fire flaps/ventilation are clear.
- Open the fuel supply valves and confirm sea suction valves are open.
- Confirm gear selector is in NEUTRAL.
- Visually inspect the engine room around each engine for fuel leaks, oil leaks, or debris.

3.3 Starting Engines

- Turn ignition key to the ON position. Allow the fuel injection system to prime (listen for fuel pump).
- Start the port engine. Allow to idle. Immediately check the raw-water overboard discharge for steady flow and the oil pressure for normal indication.
- Start the starboard engine. Allow to idle. Check the raw-water overboard discharge and oil pressure.
- If either engine does not start on the first attempt, wait 10 seconds before reattempting. If the engine fails to start after three attempts, do not continue — investigate the cause before further attempts.

3.4 Post-Start Verification

With both engines running at idle, verify the following:

- Engine run test: no strange noises, vibrations, or false starts.
- Cooling system is functioning on both engines (raw-water discharge sighted overboard; temperatures within the normal range).
- Oil pressure warning lights are extinguished.
- Battery charging indicators show normal operation on all three groups.
- Steering responds correctly in both directions.
- Throttle and gear selectors operate smoothly through forward, neutral, and reverse.
- Engine gauges (RPM, temperature, voltage) show normal readings.

Allow the engines to warm up at idle for a minimum of 2 minutes before applying load.

4. OPERATIONAL NOTES

- Do not operate engines at sustained high RPM until engine oil temperature has reached operating range.
- Monitor engine temperatures and the raw-water discharge periodically throughout the voyage. Loss of cooling flow or a rising temperature requires immediate investigation.
- If any engine alarm sounds during operation, reduce throttle to idle, assess the situation, and shut down the affected engine if necessary.
- Operate both engines at matching RPM to maintain balanced thrust and steering.
- Monitor battery voltage on all three groups periodically. If a group is not charging, investigate the charging circuit before the next voyage.

5. SHUT-DOWN PROCEDURE

5.1 At Berth or Mooring

- Reduce throttle to idle on both engines.
- Place both gear selectors in NEUTRAL.
- Allow engines to idle for 1–2 minutes to allow cooling.
- Turn ignition keys to OFF for both engines.
- Close the fuel supply and sea suction valves if the vessel will be left unattended for an extended period.

5.2 Battery Isolation

If the vessel will be unattended:

- Confirm the emergency parallel isolator is OFF.
- Switch the Port, Starboard, and House battery group isolators to OFF.

If the vessel will be attended or on a short stop, the House group isolator may remain ON at the master's discretion.

5.3 Securing the Vessel

- Confirm the engine room is secured and ventilation closures are set as required.
- Close all hatches, lockers, and access points.
- Remove or secure all loose equipment.
- Deploy fenders and mooring lines as appropriate (refer FUS-SMS-ROH-05C).
- Check bilge is dry. If water is present, identify the source before leaving the vessel.

6. RECORDS

Engine start-up and shut-down checks are captured as part of the QR code pre-start form and the physical vessel logbook. Any defects or anomalies identified during start-up or shut-down shall be recorded and reported to the East Coast Operations Superintendent.

7. REFERENCES

- FUS-SMS-ROH-05A Pre-Departure Checks
- FUS-SMS-ROH-07 Maintenance
- Detroit Diesel 6V-92T Owner's Manual
- NSCV C5A Table 5 (Bilge Pumps)
- Marine Order 504 (2024), Schedule 1, Clause 11(5)(a)